WORKED SOLUTION

STEP III 2016, Question 8

Pure • Functional Equations

STEP III 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q8

The key idea

Each equation links the unknown function at x to its value at some transformed point. If applying the transformation again (or a third time) brings you back to x, then substituting the transformed point into the equation gives a *second* equation in the same unknowns. You now have simultaneous linear equations, and you can eliminate the values you don't want, just as you would for ordinary unknowns.

Solution

Part (i)

The equation holds for *all* x, so it holds with x replaced by $-x$:

$$\mathrm{f}(-x) + (1+x)\,\mathrm{f}(x) = (-x)^2 = x^2,$$

which is the required result.

Now we have two equations in the two unknowns $\mathrm{f}(x)$ and $\mathrm{f}(-x)$. The second gives $\mathrm{f}(-x) = x^2 - (1+x)\mathrm{f}(x)$, and substituting this into the original equation gives

$$\begin{aligned} \mathrm{f}(x) + (1-x)(x^2 - (1+x)\mathrm{f}(x)) &= x^2 \\ \mathrm{f}(x)(1 - (1-x^2)) &= x^2 - (1-x)x^2 \\ x^2\,\mathrm{f}(x) &= x^3. \end{aligned}$$

For $x \neq 0$ we may divide by x^2 to get $\mathrm{f}(x) = x$. At $x = 0$ the original equation reads $\mathrm{f}(0) + \mathrm{f}(0) = 0$, so $\mathrm{f}(0) = 0$ as well. Hence

$$\mathrm{f}(x) = x \quad \text{for all } x.$$

Verification: $x + (1-x)(-x) = x - x + x^2 = x^2$, as required.

Ask yourself Why does the question insist that we verify the answer?

Our argument shows only that *if* a function satisfies the equation, it must be $\mathrm{f}(x) = x$. It doesn't show that such a function exists: the manipulations might have thrown away information, and then no function would work. Substituting back closes the gap.

Part (ii)

First note that $\mathrm{K}(x) \neq 1$ for every $x \neq 1$, since $\frac{x+1}{x-1} = 1$ would need $x+1 = x-1$. So $\mathrm{K}(\mathrm{K}(x))$ is defined, and multiplying top and bottom by $x-1$,

$$\mathrm{K}(\mathrm{K}(x)) = \frac{\dfrac{x+1}{x-1} + 1}{\dfrac{x+1}{x-1} - 1} = \frac{(x+1)+(x-1)}{(x+1)-(x-1)} = \frac{2x}{2} = x.$$

The equation for g is $\mathrm{g}(x) + x\,\mathrm{g}(\mathrm{K}(x)) = x$. Because $\mathrm{K}(x) \neq 1$, we may replace x by $\mathrm{K}(x)$, and since $\mathrm{K}(\mathrm{K}(x)) = x$ this gives

$$\mathrm{g}(\mathrm{K}(x)) + \mathrm{K}(x)\,\mathrm{g}(x) = \mathrm{K}(x).$$

Write K for $\mathrm{K}(x)$. Then $\mathrm{g}(K) = K - K\,\mathrm{g}(x)$, and substituting into the original equation,

$$\mathrm{g}(x) + x(K - K\,\mathrm{g}(x)) = x \quad \Longrightarrow \quad (1 - xK)\,\mathrm{g}(x) = x - xK.$$

Now

$$1 - xK = \frac{(x-1) - x(x+1)}{x-1} = -\frac{x^2+1}{x-1},$$
$$x - xK = x \cdot \frac{(x-1)-(x+1)}{x-1} = -\frac{2x}{x-1}.$$

Since $x^2 + 1 > 0$, the factor $1 - xK$ is never zero, so we can divide:

$$\mathrm{g}(x) = \frac{-2x/(x-1)}{-(x^2+1)/(x-1)} = \frac{2x}{x^2+1} \qquad (x \neq 1),$$

as required. (One can check that this function does satisfy the equation; for instance at $x = 2$, $\mathrm{K}(2) = 3$ and $\frac{4}{5} + 2 \cdot \frac{6}{10} = 2$.)

Part (iii)

Let $\mathrm{M}(x) = \dfrac{1}{1-x}$. The equation is

$$\mathrm{h}(x) + \mathrm{h}(\mathrm{M}(x)) = 1 - x - \mathrm{M}(x),$$

and the right-hand side is written in terms of x and $\mathrm{M}(x)$ only, which is encouraging.

This time M is not its own inverse, so look at repeated applications. For $x \neq 0, 1$:

$$\mathrm{M}(\mathrm{M}(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}, \qquad \mathrm{M}(\mathrm{M}(\mathrm{M}(x))) = \frac{1}{1 - \frac{x-1}{x}} = \frac{x}{1} = x.$$

So M has period three. We also need M to keep us inside the domain: $\mathrm{M}(x)$ is never 0, and $\mathrm{M}(x) = 1$ only when $x = 0$, so if $x \neq 0, 1$ then $\mathrm{M}(x) \neq 0, 1$ too.

Ask yourself Two applications of M don't return to x. How do we get enough equations?

Use all three points of the cycle. Write $a = x$, $b = \mathrm{M}(x)$, $c = \mathrm{M}(b) = \frac{x-1}{x}$, so that $\mathrm{M}(c) = a$. Applying the equation at a, b and c in turn:

$$\begin{aligned} \mathrm{h}(a) + \mathrm{h}(b) &= 1 - a - b, \\ \mathrm{h}(b) + \mathrm{h}(c) &= 1 - b - c, \\ \mathrm{h}(c) + \mathrm{h}(a) &= 1 - c - a. \end{aligned}$$

Adding all three and halving,

$$\mathrm{h}(a) + \mathrm{h}(b) + \mathrm{h}(c) = \tfrac{3}{2} - (a + b + c).$$

Subtracting the second equation leaves

$$\mathrm{h}(a) = \tfrac{3}{2} - a - b - c - (1 - b - c) = \tfrac{1}{2} - a.$$

Hence

$$\mathrm{h}(x) = \tfrac{1}{2} - x \qquad (x \neq 0,\ x \neq 1).$$

Verification: $(\frac{1}{2} - x) + (\frac{1}{2} - \frac{1}{1-x}) = 1 - x - \frac{1}{1-x}$, as required.

Checks

- In (i), try $x = 3$: $3 + (-2)(-3) = 9 = 3^2$.
- In (ii), try $x = -1$: $\mathrm{K}(-1) = 0$, and $\mathrm{g}(-1) + (-1)\,\mathrm{g}(0) = -1 - 0 = -1$. Also $\mathrm{K}(0) = -1$, confirming the period two.
- In (iii), try $x = 2$: the cycle is $2 \to -1 \to \frac{1}{2} \to 2$, and $\mathrm{h}(2) + \mathrm{h}(-1) = -\frac{3}{2} + \frac{3}{2} = 0$, while $1 - 2 - (-1) = 0$.

All three functions were also confirmed by a computer check of each equation at two thousand random rational values, using exact arithmetic.

Common slips

- Dividing by x^2 in (i) and forgetting $x = 0$, which needs its own (one-line) argument.

- Substituting $\mathrm{K}(x)$ or $\mathrm{M}(x)$ for x without checking that it stays in the domain where the equation holds.
- In (iii), substituting only once and getting stuck with three unknowns: you need the whole cycle.

Beyond STEP

The maps $x \mapsto -x$, K and M are all Möbius transformations $x \mapsto \frac{ax+b}{cx+d}$, and these form a group under composition. A transformation of finite order generates a finite cyclic group, and the method here is to sum (or eliminate) over that group, an idea that reappears in group representation theory. Möbius maps correspond to 2×2 matrices, and you can find $\mathrm{M}^3 =$ identity by cubing a matrix. Needham's *Visual Complex Analysis* has a beautiful treatment of Möbius transformations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.